Wave Speed

A wave of wavelength λ and frequency f travels at

$$v = \lambda f.$$

The wave number is $k = 2\pi/\lambda$.

Beat Frequency

- Two tones with frequencies f_1 and f_2 are added.
- The beat frequency is $f_b = |f_1 - f_2|$.

$$\cos a + \cos b = 2 \cos \frac{a+b}{2} \cos \frac{a-b}{2}.$$